ES-FA: Event-Driven SNN Acceleration with Local Software-Hardware Evidence

BTP Progress Draft

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Abstract

This draft summarizes the current ES-FA implementation after the latest local smoke run. The software pipeline ran successfully with baseline training, hardware-aware loss, adaptive dataflow, analysis, evidence tables, and output section generation. The best current software result is `exp1_hardware_aware_loss`, which keeps 95.70% validation accuracy while reducing the energy proxy by 79.68% compared with the dense baseline estimate. Fresh hardware/Vivado execution was skipped in this run; existing local hardware-validation rows remain available for reference, while KV260 board measurements are still pending.

1 Motivation

Edge FPGA inference needs models that are not only accurate but also efficient. Spiking neural networks are useful for this direction because sparse spike activity can potentially reduce memory traffic and compute operations. ES-FA uses a small SNN and tracks both ML metrics and hardware-facing proxy metrics.

2 Current Implementation

The repository is organized around the renamed project layout:

- `p1_training/`: SNN model, training core, baseline runner, exports.
- `p2_hardware_model/`: spike, memory, energy, and latency estimators.
- `experiments/`: hardware-aware and adaptive dataflow experiments.
- `analysis/`: ranking, plots, and evidence generation.
- `hardware/`: Verilog RTL blocks and testbenches.
- `hardware_validation/`: KV260-oriented xsim/Vivado validation flow.
- `deployment/` and `system_layer/`: CLI and adaptive routing.
- `output/`: generated report and presentation material.

3 Model and Estimator

The current SNN is a compact feed-forward model:

$$784 \to 128 \to 64 \to 10$$

with LIF hidden neurons and a default temporal window of 16 time steps. During training, ES-FA records spike activity, sparsity, memory-access estimates, energy proxy, and latency proxy.

4 Latest Smoke-Run Results

Run	Accuracy	Sparsity	Energy proxy	Score
`exp1_hardware_aware_loss`	0.9570	0.5648	4,592,863.17	0.7699
`baseline_paper1`	0.9570	0.5648	22,604,295.76	0.7387
`exp5_dataflow_adaptation`	0.9570	0.5648	45,016,894.73	0.6699

Table 1: Current software ranking from the latest local smoke run.

The current best run is `exp1_hardware_aware_loss`. It matches the baseline accuracy and sparsity in this smoke run, but the energy proxy is much lower. The proxy reduction compared with the dense baseline is:

$$\frac{4{,}592{,}863.17 - 22{,}604{,}295.76}{22{,}604{,}295.76} = -79.68\%.$$

5 Hardware Status

The hardware side contains RTL blocks and a KV260-centered validation flow. The latest smoke run used `-SkipVivado -SkipHardware`, so no fresh hardware build was executed during this run. Existing local evidence rows still include simulation/report data such as 5760 ns active-window latency and 576 cycles, but these should be described as previous local validation evidence, not a new board measurement.

6 Presentation Claim Boundary

The safe claim for the current presentation is:

> The local software and analysis pipeline is working end-to-end. The best hardware-aware software experiment preserves 95.70% accuracy and reduces the energy proxy by 79.68% against the dense baseline estimate. Hardware RTL and validation scripts are present, but fresh Vivado hardware execution and KV260 physical measurements remain pending for final hardware claims.

7 Next Work

- Run xsim-only validation after the presentation smoke run.
- Run full Vivado validation when build time is available.

- Collect physical KV260 board measurements.
- Calibrate estimator constants against measured hardware values.
- Expand the model/scheduler validation matrix.

8 Conclusion

ES-FA is now presentation-ready as a local software-hardware research pipeline. The project can demonstrate training, hardware-aware metrics, analysis plots, evidence tables, and CLI routing. The remaining work is mainly hardware closure: fresh Vivado runs, KV260 board measurements, and estimator calibration.